

Modelling Report

PMP Solutions – AI Fraud Detection System

1. Exploratory Data Analysis (EDA)

Before model training, the dataset was explored to understand its structure, quality and fraud patterns. We examined dataset dimensions, data types, summary statistics, missing values, duplicate records and the distribution of the target variable. We analysed transaction types, transaction amounts, sender and receiver balances and the fraud label. Visualisations included fraud versus legitimate transactions, transaction type distribution, fraud cases by transaction type, transaction amount distribution and feature importance. The analysis showed that fraud is rare and concentrated mainly in TRANSFER and CASH_OUT transactions.

2. Data Cleaning

The dataset was checked for missing values, duplicate rows, invalid data types, zero-value transactions and negative transaction amounts. Cleaning improves data quality, reduces bias and increases model reliability.

3. Class Imbalance Strategy

Fraud cases represented only a small percentage of the dataset. To prevent the model from favouring the majority class, class weighting was applied during training. Performance was evaluated using Precision, Recall, F1-Score and ROC-AUC instead of relying only on Accuracy.

4. Feature Engineering

New features included BalanceChangeOrig, BalanceChangeDest, BalanceChangeRatio, HighValueTransaction, AccountEmptied and NewDestinationAccount. These features capture behavioural patterns that are strongly associated with fraudulent activity.

5. Feature Selection

Account identifiers were excluded because they do not generalise to unseen data and can lead to overfitting. The final feature set focused on transaction behaviour and financial balances.

6. Model Development

Logistic Regression, Decision Tree, Random Forest and XGBoost models were trained and compared using the engineered features. Hyperparameters were selected to improve generalisation while limiting overfitting.

7. Model Evaluation

Models were assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC and Confusion Matrices. Recall was prioritised because detecting fraudulent transactions is critical in financial systems.

8. Explainability

SHAP was used to explain predictions by identifying the features contributing most to fraud classifications, improving transparency and trust.

9. Conclusion

Combining EDA, data cleaning, imbalance handling, feature engineering, feature selection and model comparison produced a robust fraud detection solution suitable for integration into an interactive dashboard.